

Revisiting the Mechanical Advantage of Compliant Mechanisms

iNaCoMM 2025

R. Vishwanath G. K. Ananthasuresh

Multidisciplinary and Multiscale Design and Device (**M2D2**) Laboratory,
Indian Institute of Science, Bengaluru

December 08, 2025



1. Introduction

2. Mechanical advantage based on kinetoelastics

3. Comparing rigid-body and compliant mechanisms

4. On Prof. Midha's famous compliant crimper

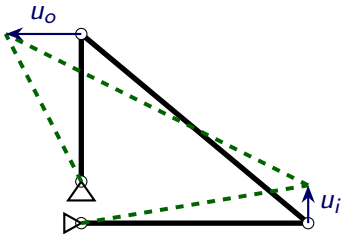
Introduction



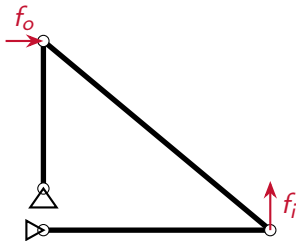
For single-input single-output mechanisms,

- **Displacement advantage** (DA) refers to the ratio of the displacement of the designated output and input degrees of freedom,
- **Mechanical advantage** (MA) refers to the ratio of force experienced along the designated output and input degrees of freedom.

$$DA = \frac{u_o}{u_i}$$



$$MA = \frac{f_o}{f_i}$$



Revisiting MA in compliant mechanisms



In Salamon and Midha (1998)*:

Initial state:

$$f_o = f_o^*$$

$$f_i = f_i^*$$

$$u_i = u_i^*$$

$$u_o = u_o^*$$

$$SE = SE^*$$

Perturbed state:

$$f_o = f_o^* + \Delta f_o^*$$

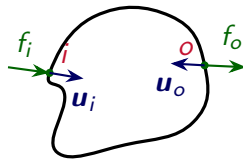
$$f_i = f_i^* + \Delta f_i^*$$

$$u_i = u_i^* + \Delta u_i^*$$

$$u_o = u_o^* + \Delta u_o^*$$

$$SE = SE^* + \Delta SE^*$$

$$\Delta SE^* = f_i^* \Delta u_i^* - f_o^* \Delta u_o^*$$



$$MA = \frac{f_o^*}{f_i^*} = \frac{\Delta u_i^*}{\Delta u_o^*} - \frac{\Delta SE^*}{f_i^* \Delta u_o^*} \equiv MA_r - MA_c$$

*B. A. Salamon and A. Midha. "An Introduction to Mechanical Advantage in Compliant Mechanisms".
In: *Journal of Mechanical Design* 120.2 (June 1998), pp. 311–315.

Revisiting MA in compliant mechanisms



$$\text{MA} = \frac{\Delta u_i^*}{\Delta u_o^*} - \frac{\Delta SE^*}{f_i^* \Delta u_o^*} \equiv \text{MA}_r - \text{MA}_c$$

Excerpt from Salamon and Midha (1998):

*“The first term in Eq. (14) takes the form of a **rigid-body mechanical advantage**. This term would result if an **instant center analysis** for mechanical advantage (Shigley and Uicker, 1980) could be applied to the compliant mechanism in any instantaneous position...”*

Revisiting MA in compliant mechanisms

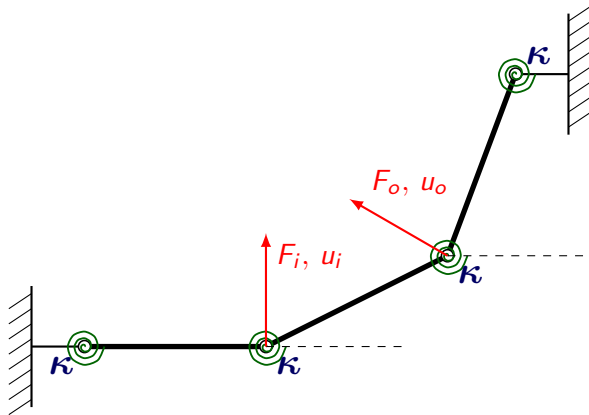


$$\text{MA} = \frac{\Delta u_i^*}{\Delta u_o^*} - \frac{\Delta SE^*}{f_i^* \Delta u_o^*} \equiv \text{MA}_r - \text{MA}_c$$

*“The second term in Eq. (14) also resembles a mechanical advantage term. It is referred to as the **compliant component of the mechanical advantage** (MA_c), and it accounts for the energy stored in the mechanism...”*



1. Introduction
- 2. Mechanical advantage based on kinetoelastics**
3. Comparing rigid-body and compliant mechanisms
4. On Prof. Midha's famous compliant crimper



On κ

$\kappa \rightarrow 0 \implies$ 4-bar rigid-body linkage

$\kappa \rightarrow \infty \implies$ compliant 4-bar linkage



$$[K_t(\mathbf{u})]\{\Delta \mathbf{u}\} = \{\Delta \mathbf{f}\} \iff \{\Delta \mathbf{u}\} = [C_t(\mathbf{u})]\{\Delta \mathbf{f}\}$$

For a single-input single-output mechanism:

$$\begin{Bmatrix} \Delta u_i \\ \Delta u_o \\ \Delta \mathbf{u}_k \end{Bmatrix} = \begin{bmatrix} C_{ii} & C_{io} & \mathbf{C}_{ik} \\ C_{oi} & C_{oo} & \mathbf{C}_{ok} \\ \mathbf{C}_{ki} & \mathbf{C}_{ko} & \mathbf{C}_{kk} \end{bmatrix} \begin{Bmatrix} \Delta f_i \\ -\Delta f_o \\ \cancel{\Delta \mathbf{f}_k} \end{Bmatrix}$$

$$\Delta u_i = [C_t(\mathbf{u})]_{ii} \Delta f_i - [C_t(\mathbf{u})]_{io} \Delta f_o$$

$$\Delta u_o = [C_t(\mathbf{u})]_{oi} \Delta f_i - [C_t(\mathbf{u})]_{oo} \Delta f_o$$



$$\Delta u_o = [C_t(\mathbf{u})]_{oi} \Delta f_i - [C_t(\mathbf{u})]_{oo} \Delta f_o$$
$$\Delta f_o = \frac{[C_t(\mathbf{u})]_{oi} \Delta f_i}{[C_t(\mathbf{u})]_{oo}} - \frac{\Delta u_o}{[C_t(\mathbf{u})]_{oo}}$$

Integrating the last expression from **initial state** ($1 \equiv \{u_i = 0, u_o = 0, f_i = 0, f_o = 0\}$) to its **final state** ($2 \equiv \{u_i = u_i^*, u_o = u_o^*, f_i = f_i^*, f_o = f_o^*\}$):

$$f_o^* = \int_1^2 \frac{[C_t(\mathbf{u})]_{oi} df_i}{[C_t(\mathbf{u})]_{oo}} - \int_1^2 \frac{du_o}{[C_t(\mathbf{u})]_{oo}}$$



The last expression can be rewritten in the following form:

$$\frac{f_o^*}{f_i^*} = \text{MA}_s \left(1 - \frac{f_n}{f_i^*} \right)$$
$$f_n \equiv \left(\frac{\int_1^2 \frac{du_o}{[C_t(\mathbf{u})]_{oo}}}{\frac{1}{f_i^*} \int_1^2 \frac{[C_t(\mathbf{u})]_{oi} df_i}{[C_t(\mathbf{u})]_{oo}}} \right), \quad \text{MA}_s = \frac{1}{f_i^*} \int_1^2 \frac{[C_t(\mathbf{u})]_{oi} df_i}{[C_t(\mathbf{u})]_{oo}}$$

f_n is strictly non-negative, and zero only when $u_o^* = 0$.

Thus, **MA is maximized when the mechanism behaves like a structure**. Consequently, $\text{MA}_s \times f_n/f_i^*$ can be interpreted as the loss in MA necessary to engage the workpiece.

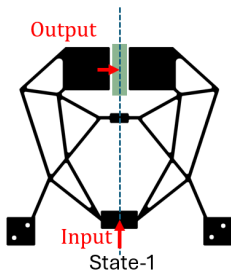
MA based on kinetoelastics



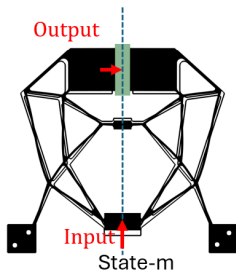
Let us define the intermediate stage where the mechanism begins to engage the workpiece as **state-m** $\equiv \{u_i^m, u_o^*, f_i^m, f_o = 0\}$.

$$\frac{f_o^*}{f_i^*} = \frac{1}{f_i^*} \int_1^2 \frac{[C_t(\mathbf{u})]_{oi} df_i}{[C_t(\mathbf{u})]_{oo}} - \frac{1}{f_i^*} \int_1^m \frac{[C_t(\mathbf{u})]_{oi} df_i}{[C_t(\mathbf{u})]_{oo}} = \frac{1}{f_i^*} \int_{f_i^m}^{f_i^*} \frac{[C_t(\mathbf{u})]_{oi} dF_i}{[C_t(\mathbf{u})]_{oo}},$$

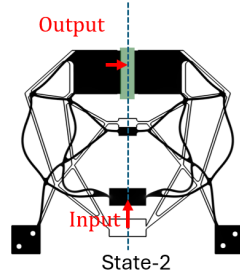
where f_i^m is the input force at which the mechanism engages the workpiece.



$$\{f_i = 0, u_i = 0, f_o = 0, u_o = 0\}$$



$$\{f_i = f_i^m, u_i = u_i^m, f_o = 0, u_o = u_o^*\}$$



$$\{f_i = f_i^*, u_i = u_i^2, f_o = f_o^2, u_o = u_o^*\}$$

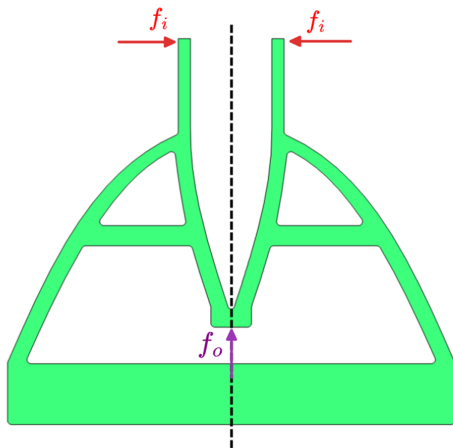
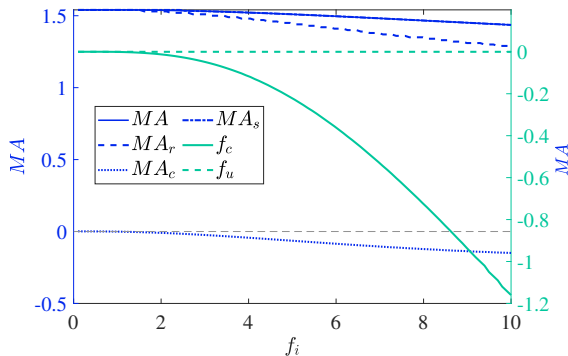
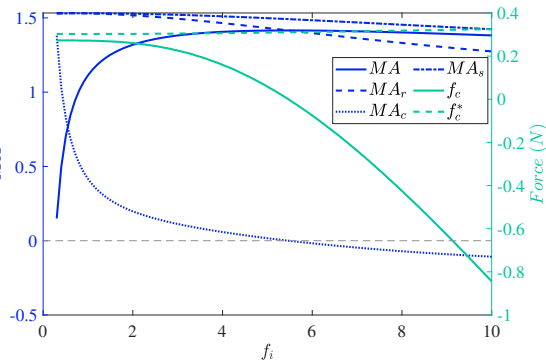


Figure: A compliant crimping mechanism

MA based on kinetoelastics



(a) Variability of the different MA terms and compliance forces with input force for $u_o^* = 0$



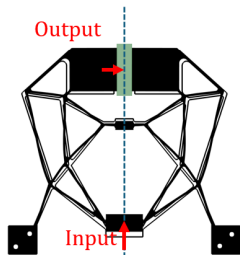
(b) Variability of the different MA terms and compliance forces with input force for $u_o^* = 2$ mm

MA based on kinetoelastics

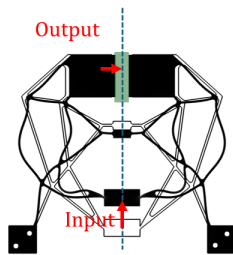


$$MA = MA_r - MA_c = MA_s - MA_n$$

$$MA = \frac{\Delta u_i^*}{\Delta u_o^*} - \frac{\Delta SE^*}{f_i^* \Delta u_o^*} = \frac{1}{f_i^*} \int_1^2 \frac{[C_t(\mathbf{u})]_{oi} df_i}{[C_t(\mathbf{u})]_{oo}} - \frac{1}{f_i^*} \int_1^2 \frac{du_o}{[C_t(\mathbf{u})]_{oo}}$$



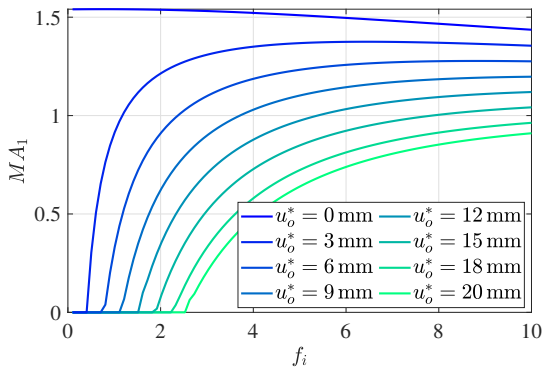
MA=0



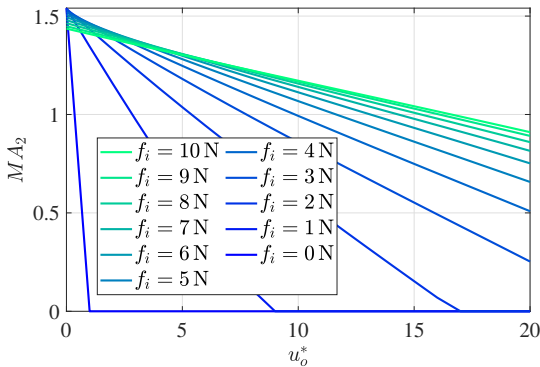
MA≠0,
MA_n^{II}=MA_n^m, MA_c^{II}≠MA_c^m

- MA_c is the loss/gain in MA due to compliance.
- MA_n is the loss in MA necessary to engage the workpiece.

$$\frac{f_o^*}{f_i^*} = \frac{1}{f_i^*} \int_1^2 \frac{[C_t(\mathbf{u})]_{oi} df_i}{[C_t(\mathbf{u})]_{oo}} - \frac{1}{f_i^*} \int_1^2 \frac{du_o}{[C_t(\mathbf{u})]_{oo}}$$



(a) MA₁



(b) MA₂



Main takeaways

1. It can be summarised that the MA in general is **maximum when the mechanism behaves like a structure** (output port is fixed),
2. There is a **loss in MA** due to the deformation the system must experience to **engage the workpiece**.
3. The **MA** at the particular load is the **average of MA_r** (the ratio of incremental displacement of input and output) **following workpiece engagement**.

Eigenvalue-based decomposition of DA and MA



$$[K]\{u\} = \{f\}$$

eigenvalues: $(\lambda_k \forall k \in \{1, 2..n\})$

orthonormal eigenvectors: $(a_k \forall k \in \{1, 2..n\})$

$$\{u\} = \sum_{k=1}^n \frac{(a_k \otimes a_k)\{f\}}{\lambda_k}$$

Where, $\otimes$ denotes dyadic product, defined as:

$$b \otimes c \equiv (b \otimes c)_{ij} = b_i \cdot c_j$$

Eigenvalue based decomposition of DA and MA



For single-input single-output mechanisms:

$$u_i = \sum_{k=1}^n \frac{f_i(a_k \otimes a_k)_{ii}}{\lambda_k} - \frac{f_o(a_k \otimes a_k)_{io}}{\lambda_k}$$
$$u_o = \sum_{k=1}^n \frac{f_i(a_k \otimes a_k)_{io}}{\lambda_k} - \frac{f_o(a_k \otimes a_k)_{oo}}{\lambda_k}$$

The last relation can be simplified to:

$$\frac{f_o}{f_i} = \text{MA}_s \left(1 - \frac{f_n}{f_i} \right)$$
$$f_n \equiv \frac{u_o}{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}, \quad \text{MA}_s = \frac{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{oo}}{\lambda_k}}$$

Eigenvalue based decomposition of DA and MA



Similarly, it is possible to represent the displacement advantage as:

$$\frac{u_o}{u_i} = \frac{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{ii}}{\lambda_k}} \left(1 - \frac{u_c}{u_i} \right)$$
$$u_c \equiv \frac{f_o \left(\left(\sum_{k=1}^n \frac{(a_k \otimes a_k)_{ii}}{\lambda_k} \right) \left(\sum_{k=1}^n \frac{(a_k \otimes a_k)_{oo}}{\lambda_k} \right) - \left(\sum_{k=1}^n \frac{(a_k \otimes a_k)_{io}}{\lambda_k} \right)^2 \right)}{\left(\sum_{k=1}^n \frac{(a_k \otimes a_k)_{io}}{\lambda_k} \right)}$$

Eigenvalue based decomposition of DA and MA



DA_f can be rewritten as:

$$DA_f = \frac{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{ii}}{\lambda_k}} = \frac{\sum_{k=1}^n \frac{(a_k)_i \times (a_k)_o}{\lambda_k}}{\sum_{k=1}^n \frac{(a_k)_i \times (a_k)_i}{\lambda_k}} = \frac{\sum_{k=1}^n \frac{(a_k)_i^2}{\lambda_k} DA_k}{\sum_{k=1}^n \frac{(a_k)_i^2}{\lambda_k}},$$

where DA_k is the ratio $(a_k)_o / (a_k)_i$ for k^{th} eigenvector. Also,

$$MA_s = \frac{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{oo}}{\lambda_k}} = \frac{\sum_{k=1}^n \frac{(a_k)_i \times (a_k)_o}{\lambda_k}}{\sum_{k=1}^n \frac{(a_k)_o \times (a_k)_o}{\lambda_k}} = \frac{\sum_{k=1}^n \frac{(a_k)_o^2}{\lambda_k} \left(\frac{1}{DA_k} \right)}{\sum_{k=1}^n \frac{(a_k)_o^2}{\lambda_k}}$$

Eigenvalue based decomposition of DA and MA



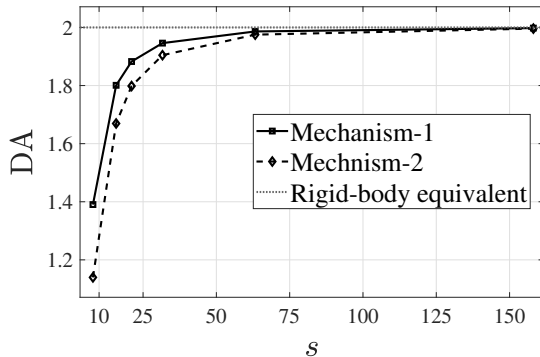
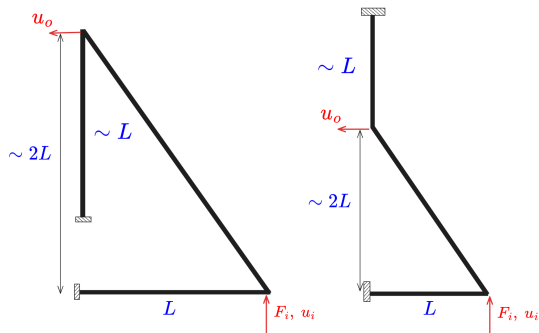
Consequently,

$$DA \times MA \leq MA_s \times DA_f = \frac{\left(\sum_{k=1}^n \frac{(a_k)_i^2}{\lambda_k} DA_k \right)^2}{\left(\sum_{k=1}^n \frac{(a_k)_i^2}{\lambda_k} DA_k^2 \right) \left(\sum_{k=1}^n \frac{(a_k)_i^2}{\lambda_k} \right)} \leq 1$$

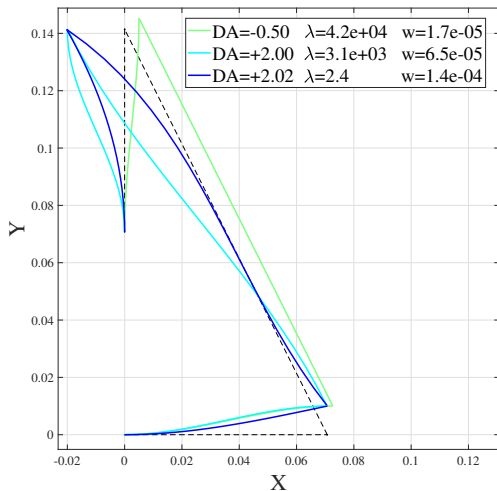
$$MA_s \times DA_f = 1$$

The equality is only satisfied when $\lambda_1 = 0$!

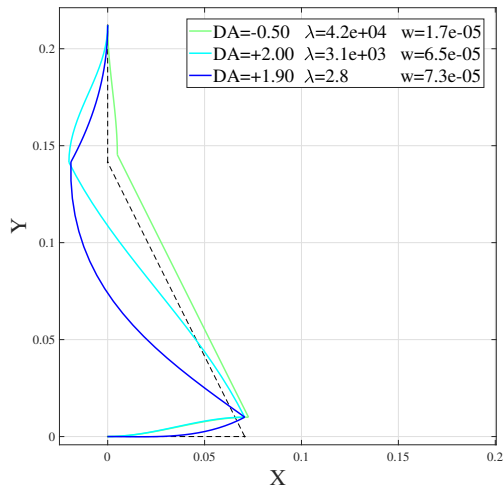
Eigenvalue based decomposition of DA and MA



Eigenvalue based decomposition of DA and MA



(a) Salient eigenvectors of mechanism-1
($s = 15.8$, $DA_f = 1.80$)



(b) Salient eigenvectors of mechanism-2
($s = 15.8$, $DA_f = 1.67$)



1. Introduction
2. Mechanical advantage based on kinetoelastics
- 3. Comparing rigid-body and compliant mechanisms**
4. On Prof. Midha's famous compliant crimper

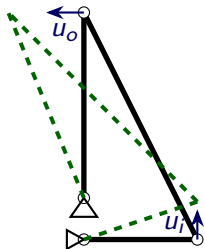
Comparing rigid-body and compliant mechanisms



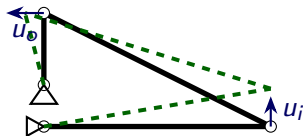
In rigid-body mechanisms, we know

$$MA = 1/DA$$

We also know that DA and MA get swapped when we flip output and input:



(a) $DA=2$, $MA=0.5$



(b) $DA=0.5$, $MA=2$

Comparing rigid-body and compliant mechanisms



Now we know,

$$DA \times MA \leq DA_f \times MA_s < 1$$

However, DA_f can be rewritten as:

$$DA_f = \frac{\sum_{k=1}^n \frac{(a_k)_i \times (a_k)_o}{\lambda_k}}{\sum_{k=1}^n \frac{(a_k)_i \times (a_k)_i}{\lambda_k}} \qquad MA_s = \frac{\sum_{k=1}^n \frac{(a_k)_i \times (a_k)_o}{\lambda_k}}{\sum_{k=1}^n \frac{(a_k)_o \times (a_k)_o}{\lambda_k}}$$

Still, the DA_f and MA_s get swapped when input and output are flipped.

Comparing rigid-body and compliant mechanisms



Uniqueness of MA and DA

- MA or DA can't be stated uniquely for a compliant mechanism!
 - ★To define MA and DA uniquely (for small deformations) for a given compliant mechanism, we must specify at least **two of the following four parameters**:
 $\{f_i, f_o, u_i, u_o\}$.
- This is also true for a rigid-body mechanism ($\lambda_1 = 0$),
 - ★MA: u_i and u_o being zero (a consequence of **static equilibrium**),
 - ★DA: f_i and f_o being zero (**mobility and degree of freedom**).

Combining compliant mechanisms



Imagine two compliant mechanisms with spring lever parameters k_i^1 , k_o^1 , n_1 and k_i^2 , k_o^2 , n_2 .

$$\begin{bmatrix} k_i^1 + (n_1)^2 k_o^1 & -n_1 k_o^1 & 0 \\ -n_1 k_o^1 & k_i^2 + (n_2)^2 k_o^2 + k_o^1 & -n_2 k_o^2 \\ 0 & -n_2 k_o^2 & k_o^2 \end{bmatrix} \begin{Bmatrix} u_i^1 \\ u_o^1 \text{ or } u_i^2 \\ u_o^2 \end{Bmatrix} = \begin{Bmatrix} f_i^1 \\ f_o^1 \text{ or } f_i^2 \\ f_o^2 \end{Bmatrix}$$

Now, eliminating the intermediate degree of freedom (no load at this point):

$$\frac{1}{k_o^2 n_2^2 + k_i^2 + k_o^1} \begin{bmatrix} k_o^1 k_o^2 n_1^2 n_2^2 + k_i^2 k_o^1 n_1^2 + k_i^2 k_o^1 n_2^2 + k_i^1 k_i^2 + k_i^1 k_o^1 & -k_o^1 k_o^2 n_1 n_2 \\ -k_o^1 k_o^2 n_1 n_2 & k_o^2 (k_i^2 + k_o^1) \end{bmatrix} \begin{Bmatrix} u_i^1 \\ u_o^2 \end{Bmatrix} = \begin{Bmatrix} f_i^1 \\ f_o^2 \end{Bmatrix}$$

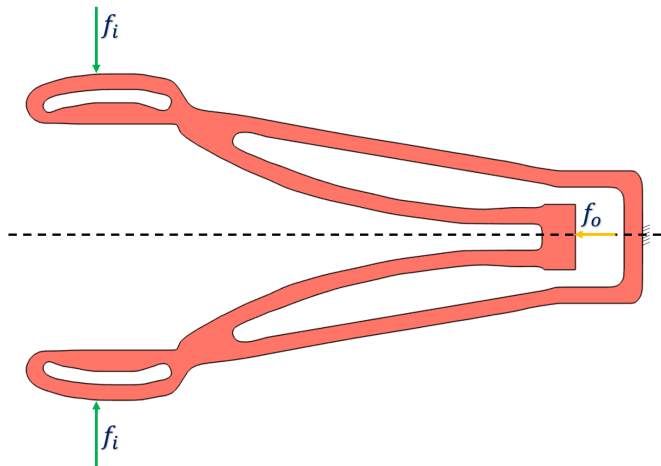
The equivalent DA for this mechanism is:

$$n^* = \left(\frac{k_o^1}{k_i^2 + k_o^1} \right) n_1 n_2$$



1. Introduction
2. Mechanical advantage based on kinetoelastics
3. Comparing rigid-body and compliant mechanisms
- 4. On Prof. Midha's famous compliant crimper**

Eigenvalue based decomposition of DA and MA



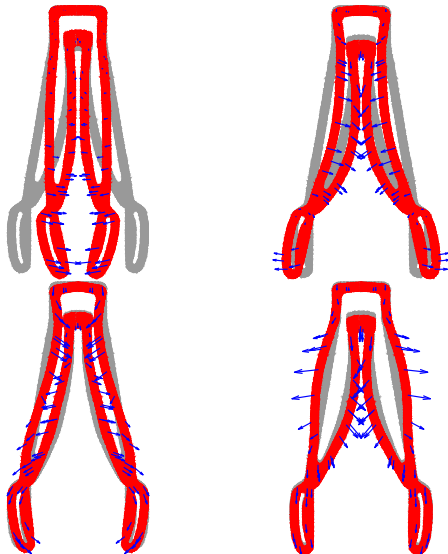
$$MA_s = 5.37,$$

$$DA_f = 0.14,$$

$$MA_s \times DA_f = 0.76$$

Figure: The first compliant mechanism designed by Prof. Ashok Midha

Eigenvalue-based decomposition of DA and MA



$$w_1 \times \text{MA}_1 = 5.21$$

$$w_2 \times \text{MA}_2 = 0.18$$

$$w_3 \times \text{MA}_3 = -0.01$$

$$w_4 \times \text{MA}_4 = -0.00$$

.

.

$$\text{Total} = 5.37$$



Thank you!